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SomeonecoolLovesMaths

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Problem

Let P and Q be on segment BC of an acute triangle ABC such that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Let M and N be the points on AP and AQ , respectively, such that P is the midpoint of AM and Q is the midpoint of AN . Prove that the intersection of BM and CN is on the circumference of triangle ABC .

Solution. Let BM intersect the circumcircle at X . Let P_∞ represent the point at infinity along AP . As a result, $(A, M; P, P_\infty) = -1 \stackrel{B}{=} (A, X; C, B)$. Similarly, if CN intersects the circle at Y then $(A, Y; C, B) = -1$. By the property of harmonic quadrilaterals, $X \equiv Y$.